

A LOW-COST TEAM-BUILDING GAME

UNDERCOOKED

Overcooked, rebuilt with real appliances, cheap RFID tags, and a laptop that keeps score — a low-cost team-building game, about \$130 for a full kitchen, where the only way to win is to work together.

Team 2b2y

SASE Hack 2026

WHO WE ARE

The Team



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MOTIVATION

Why We Built This

Overcooked is one of the best co-op games ever made — and it's still just a screen. Everyone's crowded around a TV, mashing buttons, yelling at pixels. We wanted the yelling to be about a real frying pan.

The itch — someone always said “imagine if this was real.” Eventually we stopped imagining.

The gap — no affordable kit turns a room into a physical co-op kitchen, so we built one ourselves from about \$130 of parts.

The point — a team-building game cheap enough for any club or class. Nobody cooks alone: players call out orders, hand off food, and cover for each other.

WHO IT'S FOR

Team Building Anyone Can Run

Low cost

About \$130 in parts for the whole kitchen, plus a laptop you already own.

Easy to set up

Pair each station with one touch, plug in the bridge, and open the scoreboard on any device.

Easy to play

Real controls, nothing to learn: press to chop, waggle to fry, tap a tag to plate.

Real teamwork

Nobody cooks alone: players talk, split jobs, and cover for each other under a deadline.

Built for clubs, classes, and team events.

THE CONCEPT

Overcooked, But Real

Up to eight low-cost cooking stations, cheap RFID tags standing in for ingredients, and a laptop that runs the whole game — the chaos of the video game, except you're the one holding the joystick and the frying pan, and your teammates are standing right next to you.

In the video game	In our kitchen
Press a key to chop	Press a real limit switch on a real cutting board
Waggle a stick to fry	Waggle a joystick wired straight into an ESP32
Watch a countdown on screen	Watch the same countdown on a browser everyone can see
Reset with Ctrl+R	Reset with one button — the laptop owns every bit of state

HOW IT PLAYS

One Round, Start To Finish



Progress lives on the server, not the station — pick food up mid-task, put it down on a different board of the same kind, and pick up exactly where you left off.

SYSTEM ARCHITECTURE

Four Hops, One Source Of Truth

STATIONS ≤ 8 ESP32 boards

RFID reader, LEDs, one input each

ESP-NOW

BRIDGE 1 ESP32, dumb relay

Own RFID reader, for calibration only

USB Serial

GAME SERVER Python, on a laptop

Owns every bit of game state

WebSocket

BROWSER UI Any device on Wi-Fi

Live scoreboard, 10 times a second

One function call

Reset the round and every station just draws what it's told.

CORE PRINCIPLE

All game state lives on the laptop.

Tags are dumb — cheap RFID chips, UID only — never written to.

Stations are dumb — they report events and draw whatever they're told.

The bridge is dumb — just a relay between ESP-NOW and USB serial.

Reset the whole game? One function call.

THE HARDWARE

Every Station, The Same Build

ESP32

WiFi + ESP-NOW
radio built in

MFRC522

Reads a tag's UID
over SPI — never
writes

WS2812B ×5

One wire,
addressable status
LEDs

+ one input

A limit switch, a
joystick, or nothing

≤8

stations sharing one bridge

5

LEDs per station strip

32 B

max bytes in one ESP-NOW
packet

BUDGET

Team Building On A Budget

Component	Unit cost	Qty	Notes
ESP32 dev board	~\$6	9	8 stations + 1 bridge
MFRC522 RFID reader	~\$2	9	One per station + bridge
WS2812B strip (5px)	~\$1	8	Status LEDs per station
Limit switch / joystick	~\$1.50	6	Only input-driven stations
RFID tags (13.56 MHz)	~\$0.15	30	Ingredients + plates
Wiring, perfboard, case	~\$4	9	Rough per-station estimate

~\$130

a full 8-station team-building kitchen — cheap enough for any club, class, or event — *estimate, swap in your real receipts*

MEET THE STATIONS

Six Jobs, One Shared Skeleton

Cutting Board

INPUT Limit-switch presses

RESULT Raw → Chopped

Pot

INPUT None — server timed

RESULT Cooked → Burnt

Plate

INPUT None, reader is the plate

RESULT Collects what's on it

Frying Pan

INPUT Joystick: circle or zigzag

RESULT Raw → Cooked

Sink

NEW

INPUT Joystick, any direction

RESULT Configurable wash step

Delivery

INPUT Touch a plate's tag here

RESULT Matches order, scores

TESTED IN REAL LIFE

Tested On Real Hardware

Every station, the bridge, and a full round — tested end to end on the real thing. All of it works.

✓ **Cutting Board**

Limit switch

✓ **Frying Pan**

Joystick

✓ **Pot**

Server-timed

✓ **Sink**

Joystick

✓ **Plate**

RFID tap

✓ **Delivery**

RFID tap

✓ **Bridge**

ESP-NOW ↔ USB

✓ **RFID readers**

MFRC522, every station

✓ **Status LEDs**

WS2812B strips

Talking Over ESP-NOW

STATION → BRIDGE

- Hello, Heartbeat
- TagPlaced, TagRemoved
- TaskProgress, TaskDone

BRIDGE → STATION

- Welcome, Accept
- Reject
- SetDisplay

5× retries

200ms apart

Acked and de-duplicated by (MAC, sequence) — a lost ack never runs a handler twice.

TX AABBCCDDEEFF 6 R 02

RX AABBCCDDEEFF 2 04DEADBEEF000000000000

ASCII lines, hex payloads — debuggable straight from a serial monitor.

The Server Owns Every Bit Of State

Pure Engine

A single class runs on events, a tick, and an injectable clock — no serial or network inside, so tests drive it with fake time.

One Behaviour Per Kind

A small class per station kind decides the rules; the numbers live in a level file, not in code.

Calibration Wizard

Touch any tag to the bridge to name it, touch it to every station, then touch each ingredient once. Saved to disk, reloadable.

Live Web UI

Pushes the full game state over a websocket, ten times a second, to a plain-JavaScript scoreboard — no build step.

ENGINEERING RIGOR

Built For Real Hardware

One Source Of Truth — a test parses the C++ header and fails the build the moment Python's protocol drifts from the firmware.

Zero-Hardware Dev — a simulated bridge speaks the real line protocol — the whole game runs with zero boards plugged in.

Tested End To End — Python tests plus native C++ unit tests for each station's pure logic — no ESP32 required.

102

Python tests, sub-second

21

native C++ tests, 3 firmware projects

0

boards needed to run any of them

WHAT'S NEXT

What's Cooking Next

- 01 **More levels** — new kitchens, bigger orders, and tougher recipes, so teams have to coordinate harder each round
- 02 **Better physical stations** — 3D-printed housings for the cutting board, pan, pot, and sink instead of loose breadboards, still on a budget
- 03 **More station types** — give the sink a real recipe, add new appliances, and show plate contents on their LEDs
- 04 **Bigger teams** — more players, more stations, and a full table-sized kitchen for team events

THANK YOU

Come Break It.

Jam the joystick, yank a tag mid-chop, unplug the bridge mid-round — the server is built to shrug it off and pick up right where you left it.

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